Agentic AI

Multimodal audio use-case

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Agenda

1. Training Multimodal Audio Models
2. The Architecture of Autonomous Agents
3. Hands-on workshop

Training Multimodal Audio Models

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The Core Challenge: Data Modality

Text is Discrete

LLMs are built for discrete inputs. Words are naturally distinct units. A sentence is a sequence of finite integers (tokens) from a fixed vocabulary.

Hello World

Audio is Continuous

Sound is a continuous waveform with high dimensionality. It has no natural "breaks" like spaces between words. Feeding raw waves into an LLM is impossible.

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The Tokenization Pipeline

1. Input Audio

Raw waveform (e.g., 16kHz sample rate). High data density, continuous values.

2. VQ-VAE / Codec

The "Quantizer" looks up the sound in a massive codebook. It replaces the sound with the **ID** of the closest match.

3. Audio Tokens

The output is a sequence of integers:
[45, 892, 101, 33...]
This is the "language" the LLM will learn.

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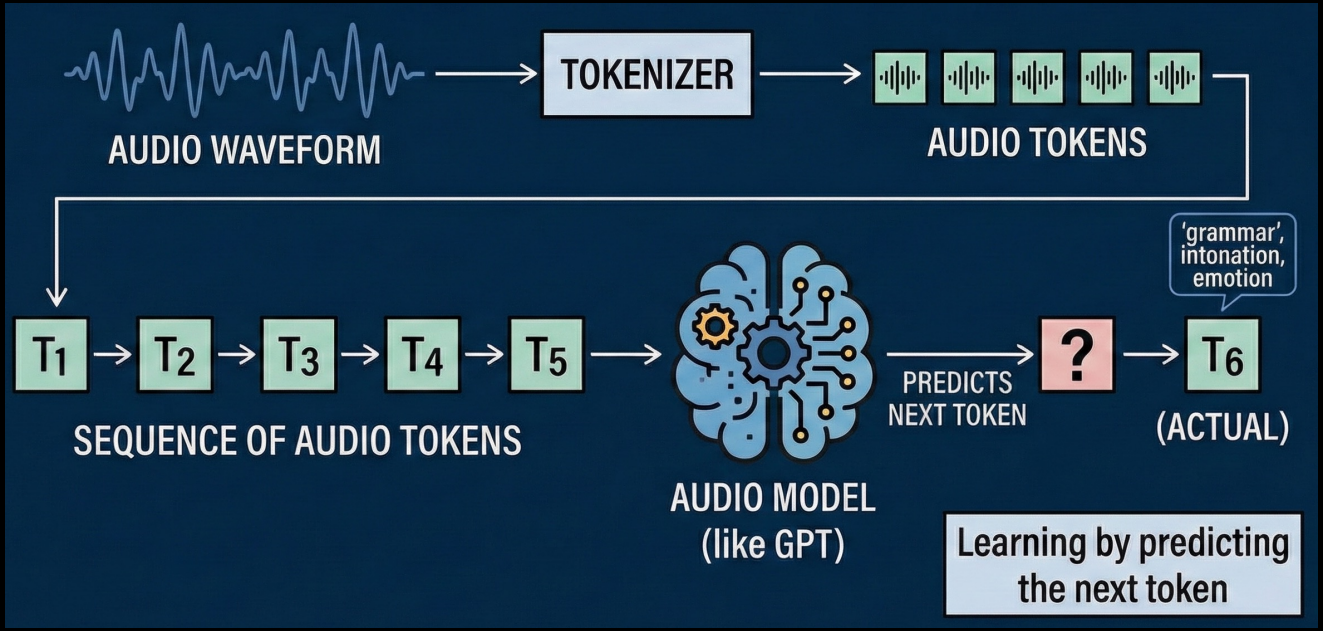
Training: Next Token Prediction

Once audio is tokenized, training is remarkably similar to text LLMs (like GPT).

The model is fed a sequence of audio tokens and learns to predict the **next token** in the sequence.

By doing this on millions of hours of audio, it learns the "grammar" of speech: phonemes, intonation, pauses, and even emotion.



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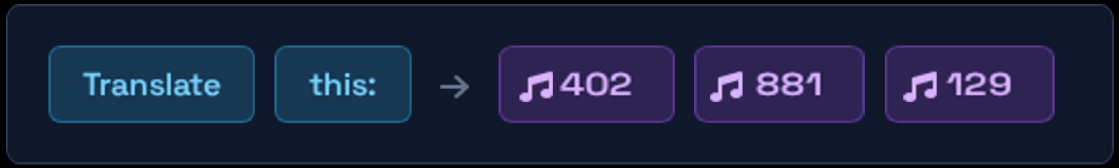
Unified Multimodal Architecture

Joint Vocabulary

State-of-the-art models use a **single vocabulary** that contains both Text Tokens and Audio Tokens.



The model reads text instructions and "generates" audio tokens as the response, mixing modalities seamlessly.

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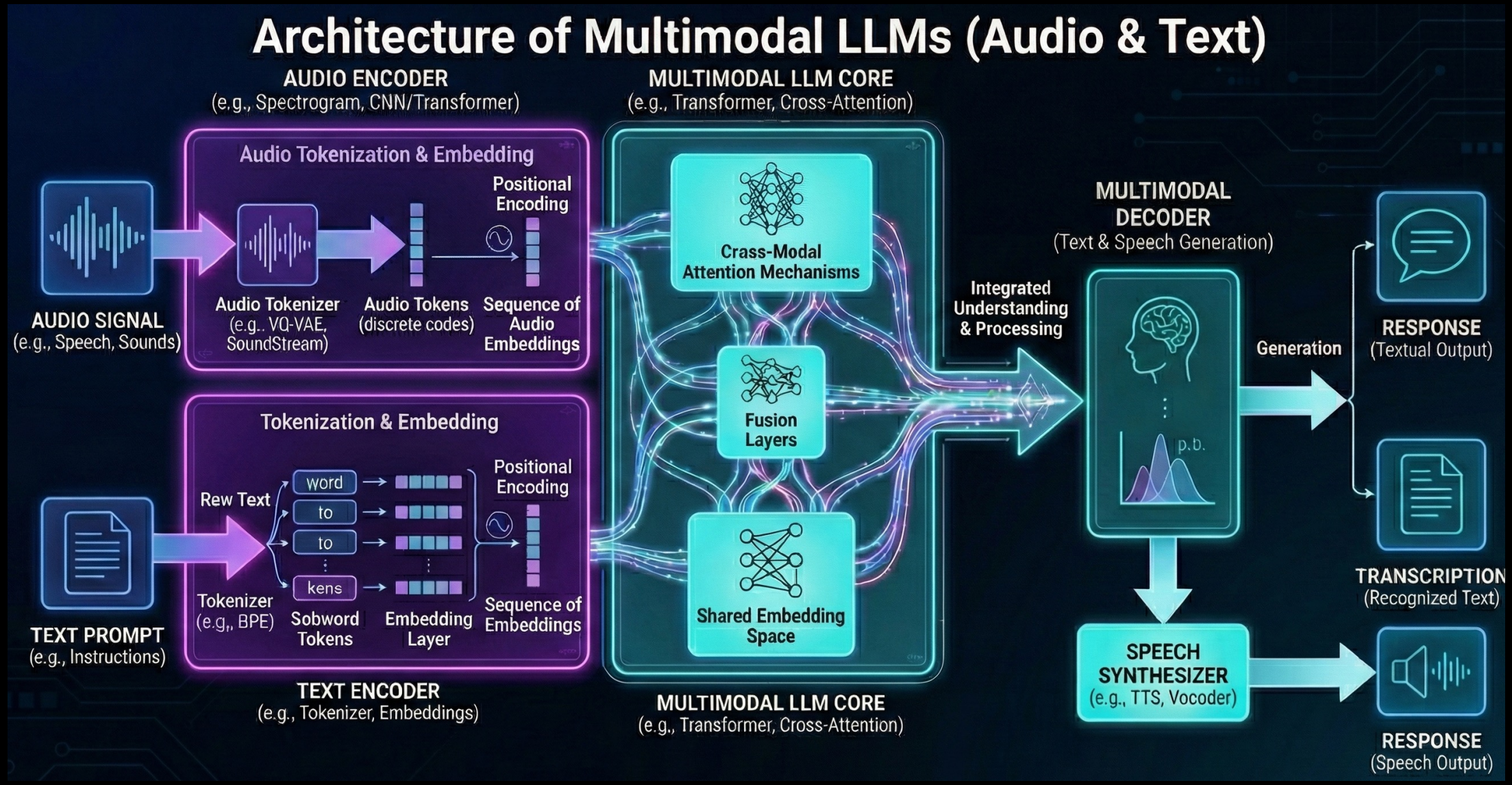

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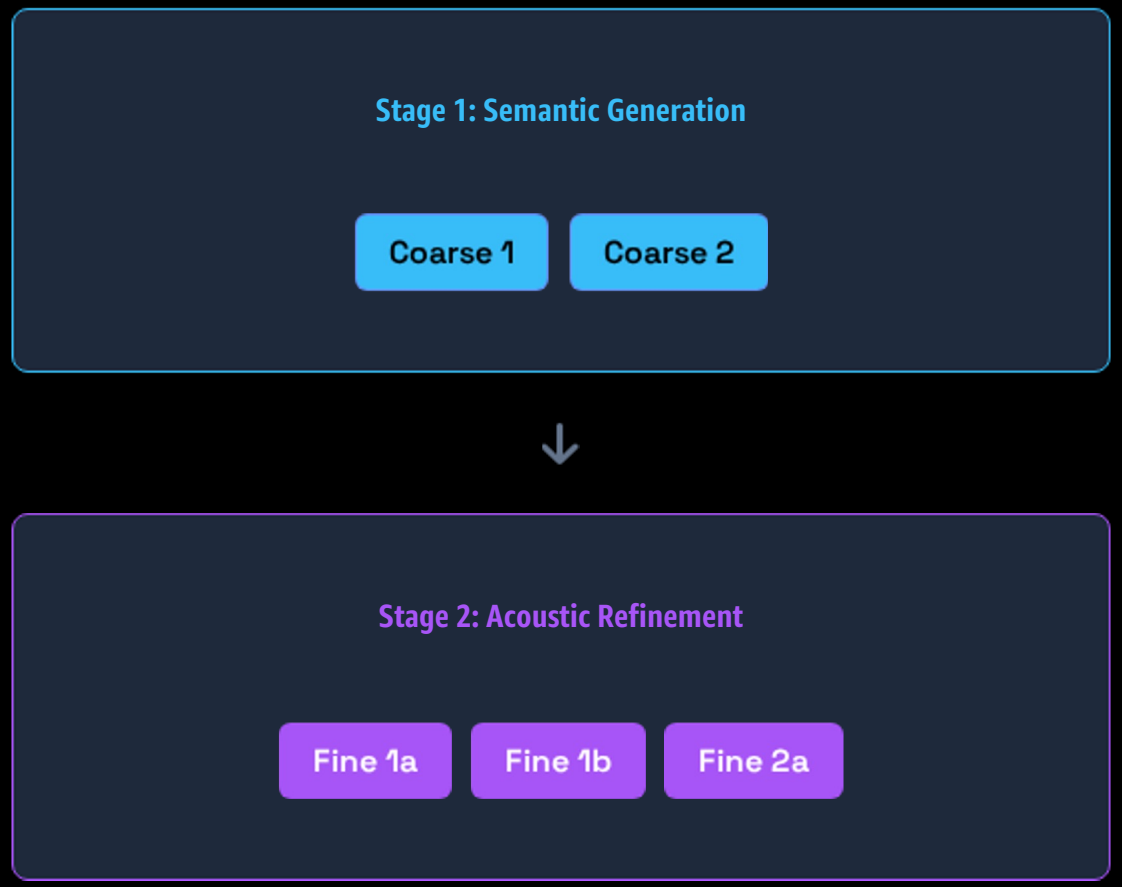
Hierarchical Tokenization

The Length Problem

Since audio sequences are so long, models often use a hierarchy of tokens to stay efficient.

- **Coarse Tokens:** Capture semantic content and prosody.
- **Fine Tokens:** Capture acoustic details (audio quality, noise).



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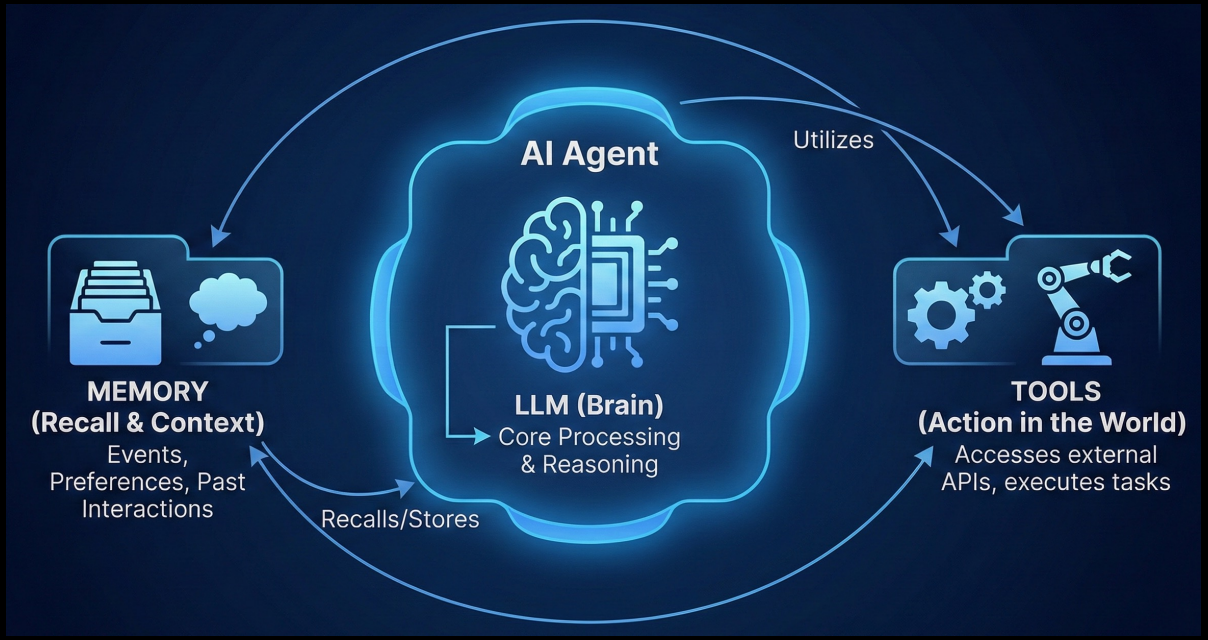
What is an AI Agent?

LLM + Agency = Agent

A standard LLM is **passive**: You give it input, it gives you output. It cannot affect the real world. An **AI Agent** is a system that uses an LLM as its "brain" to reason, but it is also equipped with:



- **Tools:** Ability to call APIs, search the web, or run code.
- **Memory:** Ability to recall events, preferences and past interactions.

  Short Term + Long Term

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The Agentic Loop - ReAct

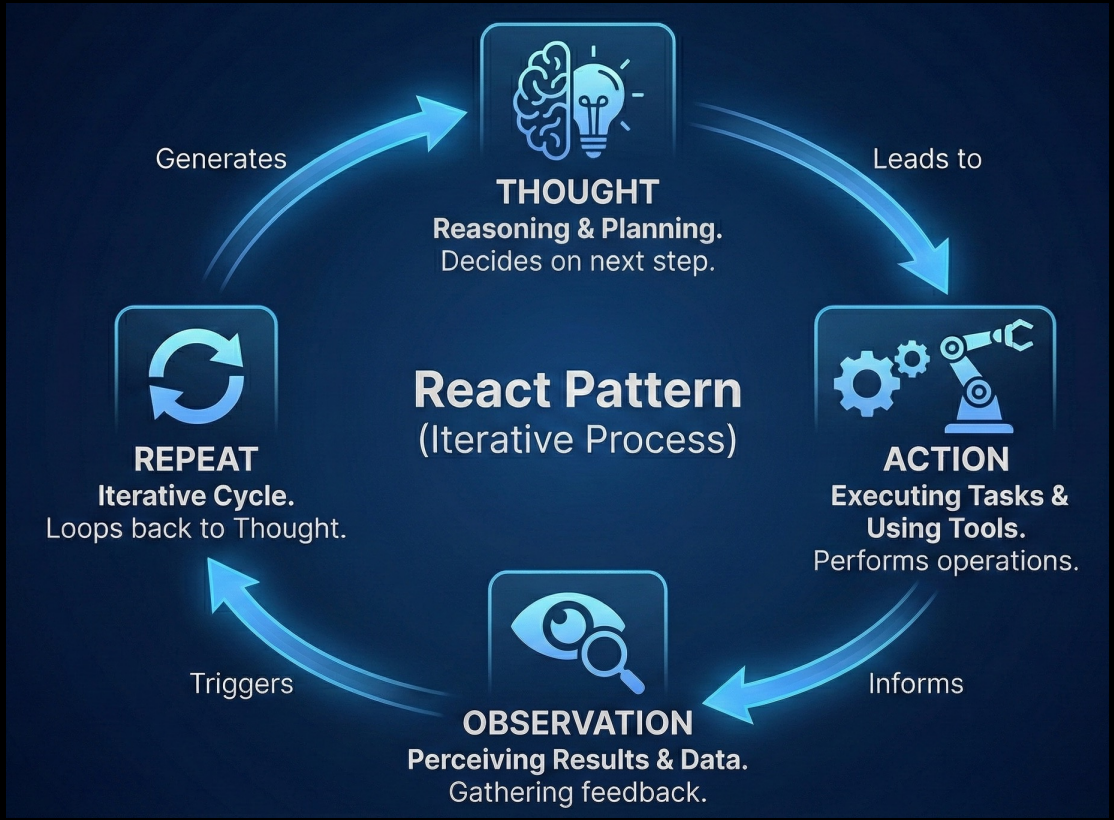
The core of an agent is the ReAct pattern (Reasoning + Acting)

1. Thought: The LLM analyzes the user request and plans the next step.

2. Action: The agent decides to call a specific tool

3. Observation: The tool executes and returns real data to the agent.

4. Repeat: The agent analyzes the new data and decides if it's finished or needs another step.



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Example: Ordering Pizza

The Goal

User: "Order a large pepperoni pizza for me."

How does the agent go from this simple text to a physical delivery?

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The Execution Trace

THOUGHT The user wants a pizza. I have the item ("pepperoni") and size ("large").

However, **I do not know the user's delivery address**. I need to find that first.

ACTION

```
call_tool: user_database.get_address(user_id="current_user")
```

OBSERVATION *Output: "123 Main Street, New York, NY"*

THOUGHT I now have the address. I have all required information to place the order.

ACTION

```
call_tool: delivery_api.order_food(
   item="pepperoni pizza",
   size="large",
   address="123 Main St"
 )
```

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Hands-On

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AWS Workshop

- Access link:
- Github repo: https://github.com/pipecat-ai/aws-workshops
- Recommended: Lab 1, Lab 2, Lab 3
- Not required: Lab 4, Lab 5

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Access keys: Setup

```
# .env file

export AWS_REGION="us-east-1"

export AWS_ACCESS_KEY_ID=”access key id"

export AWS_SECRET_ACCESS_KEY=”secret access key"

export AWS_SESSION_TOKEN=”session token
```

```python
llm = AWSNovaSonicLLMService(
    secret_access_key=os.getenv("AWS_SECRET_ACCESS_KEY"),
    access_key_id=os.getenv("AWS_ACCESS_KEY_ID"),
    session_token=os.getenv("AWS_SESSION_TOKEN"),
    region=os.getenv("AWS_REGION"),
    voice_id="tiffany",
)
```

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Thank you!

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